

# Unit 4: Image Restoration and Compression

## 4.1: Image Restoration - Introduction and Noise Models

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# Outline

- 1 Introduction to Image Restoration
- 2 Image Degradation and Restoration Process
- 3 Noise Models
  - Gaussian Noise
  - Rayleigh Noise
  - Erlang (Gamma) Noise
  - Exponential Noise
  - Uniform Noise
  - Impulse (Salt-and-Pepper) Noise
- 4 Estimation of Noise Parameters
- 5 Summary

# What is Image Restoration?

## Definition

Image restoration is the process of recovering an original image from a degraded observation by modeling the degradation and applying an inverse process.

- **Goal:** Recover the original image from its degraded version
- **Key Difference from Enhancement:**
  - Restoration: Objective process based on mathematical models
  - Enhancement: Subjective process based on human perception
- **Applications:** Medical imaging, astronomy, forensics, satellite imaging

# Image Restoration vs. Image Enhancement

## Image Enhancement

- Subjective process
- No prior knowledge required
- Improves visual appearance
- Heuristic approaches
- Example: Contrast stretching

## Image Restoration

- Objective process
- Requires degradation model
- Recovers original image
- Mathematical approaches
- Example: Deblurring

# Image Degradation Model

## General Degradation Model

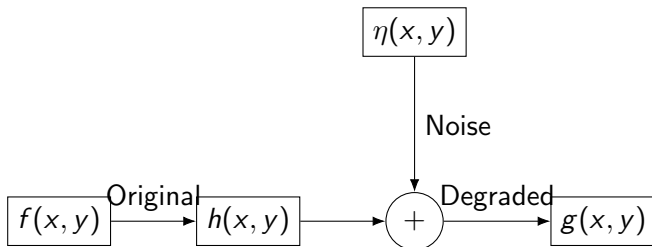
The degradation process can be modeled as:

$$g(x, y) = h(x, y) * f(x, y) + \eta(x, y)$$

where:

- $f(x, y)$  = Original image
- $h(x, y)$  = Degradation function (Point Spread Function)
- $*$  = Convolution operation
- $\eta(x, y)$  = Additive noise
- $g(x, y)$  = Degraded/Observed image

# Degradation Model - Block Diagram



**In Frequency Domain:**

$$G(u, v) = H(u, v) \cdot F(u, v) + N(u, v)$$

## Restoration Objective

Given the degraded image  $g(x, y)$  and knowledge (or estimate) of  $h(x, y)$  and  $\eta(x, y)$ , obtain an estimate  $\hat{f}(x, y)$  that is as close as possible to the original  $f(x, y)$ .

## Restoration Approaches:

- 1 **Inverse Filtering:** Direct inversion of degradation
- 2 **Wiener Filtering:** Minimize mean square error
- 3 **Constrained Least Squares:** Incorporate smoothness constraints
- 4 **Interactive Restoration:** Human-in-the-loop approaches

# Challenges in Image Restoration

- **Ill-posed Problem:**

- Solution may not exist
- Solution may not be unique
- Solution may not depend continuously on data

- **Noise Amplification:** Direct inverse filtering amplifies noise

- **Unknown Degradation:**  $h(x, y)$  may not be known exactly

- **Computational Complexity:** Large-scale matrix operations

# Introduction to Noise

## What is Noise?

Noise refers to random variations in image intensity that are not part of the true image signal.

### Sources of Noise:

- Image acquisition (sensor noise, photon noise)
- Transmission errors (channel noise)
- Quantization errors
- Environmental factors (temperature, electromagnetic interference)

### Why Study Noise Models?

- Design appropriate restoration filters
- Simulate degradation for testing algorithms
- Understand noise characteristics in different applications

# Probability Density Functions

## PDF Notation

For a random variable  $z$  representing intensity, the probability density function (PDF) is denoted as  $p(z)$  where:

$$\int_{-\infty}^{\infty} p(z) dz = 1$$

## Key Statistical Measures:

- **Mean:**  $\mu = \int_{-\infty}^{\infty} z \cdot p(z) dz$
- **Variance:**  $\sigma^2 = \int_{-\infty}^{\infty} (z - \mu)^2 \cdot p(z) dz$
- **Standard Deviation:**  $\sigma = \sqrt{\sigma^2}$

## Probability Density Function

$$p(z) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(z - \mu)^2}{2\sigma^2}\right)$$

where  $\mu$  is the mean and  $\sigma^2$  is the variance.

### Characteristics:

- Bell-shaped (Normal) distribution
- Symmetric about the mean  $\mu$
- 68% of values within  $\mu \pm \sigma$
- 95% of values within  $\mu \pm 2\sigma$
- Most common noise model in practice

## Why is Gaussian Noise Important?

- ① **Central Limit Theorem:** Sum of many random variables tends to Gaussian
- ② **Mathematical Tractability:** Easy to analyze and process
- ③ **Common in Nature:** Electronic noise, thermal noise
- ④ **Worst Case Assumption:** Provides robust restoration algorithms

## Applications:

- CCD/CMOS sensor noise
- Thermal noise in electronic circuits
- Quantization noise (for fine quantization)

# Rayleigh Noise Model

## Probability Density Function

$$p(z) = \begin{cases} \frac{2}{b}(z - a) \exp\left(-\frac{(z - a)^2}{b}\right) & \text{for } z \geq a \\ 0 & \text{for } z < a \end{cases}$$

### Parameters:

- Mean:  $\mu = a + \sqrt{\pi b/4}$
- Variance:  $\sigma^2 = \frac{b(4 - \pi)}{4}$

### Characteristics:

- Skewed distribution (not symmetric)
- Always non-negative ( $z \geq a$ )
- Used for modeling range imaging noise

# Erlang (Gamma) Noise Model

## Probability Density Function

$$p(z) = \begin{cases} \frac{a^b z^{b-1}}{(b-1)!} \exp(-az) & \text{for } z \geq 0 \\ 0 & \text{for } z < 0 \end{cases}$$

where  $a > 0$ ,  $b$  is a positive integer.

### Parameters:

- Mean:  $\mu = \frac{b}{a}$
- Variance:  $\sigma^2 = \frac{b}{a^2}$

### Special Cases:

- When  $b = 1$ : Reduces to Exponential distribution
- Laser imaging applications

# Exponential Noise Model

## Probability Density Function

$$p(z) = \begin{cases} a \exp(-az) & \text{for } z \geq 0 \\ 0 & \text{for } z < 0 \end{cases}$$

where  $a > 0$ .

### Parameters:

- Mean:  $\mu = \frac{1}{a}$
- Variance:  $\sigma^2 = \frac{1}{a^2}$

### Characteristics:

- Special case of Erlang distribution with  $b = 1$
- Memoryless property
- Models waiting times, laser imaging

# Uniform Noise Model

## Probability Density Function

$$p(z) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq z \leq b \\ 0 & \text{otherwise} \end{cases}$$

### Parameters:

- Mean:  $\mu = \frac{a+b}{2}$
- Variance:  $\sigma^2 = \frac{(b-a)^2}{12}$

### Characteristics:

- Constant probability between  $a$  and  $b$
- All values equally likely in the range
- Used in quantization studies
- Simple mathematical model for analysis

## Probability Density Function

$$p(z) = \begin{cases} P_a & \text{for } z = a \\ P_b & \text{for } z = b \\ 0 & \text{otherwise} \end{cases}$$

where  $P_a$  and  $P_b$  are probabilities, and typically  $a < b$ .

### Characteristics:

- Also called "Salt-and-Pepper" noise
- $a$  appears as dark spots (pepper),  $b$  appears as bright spots (salt)
- **Bipolar:** Both  $a$  and  $b$  present (usually  $P_a \approx P_b$ )
- **Unipolar:** Either  $P_a = 0$  or  $P_b = 0$

# Impulse Noise - Properties

## Common Sources:

- Faulty memory locations in hardware
- Bit errors in transmission
- Timing errors in analog-to-digital conversion
- Malfunctioning sensor pixels (stuck pixels)

## Restoration Approach:

- Median filtering is very effective
- Linear filters (e.g., mean filter) are less effective
- Can detect and replace corrupted pixels

## Note

Unlike Gaussian noise, impulse noise corrupts only a small percentage of pixels but with very large values.

# Why Estimate Noise Parameters?

## Importance:

- Design optimal restoration filters (e.g., Wiener filter)
- Evaluate image quality
- Adaptive noise filtering
- System characterization and calibration

## Two Main Approaches:

- 1 **From Image Strips:** Using uniform regions in the image
- 2 **From Histogram:** Analyzing the intensity distribution

# Method 1: Estimation from Image Strips

## Procedure:

- 1 Select a relatively uniform region in the degraded image
- 2 Assume this region should have constant intensity in the original
- 3 Compute statistics from this region

## Formulas:

- Mean:  $\hat{\mu} = \frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N g(i,j)$
- Variance:  $\hat{\sigma}^2 = \frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N [g(i,j) - \hat{\mu}]^2$

where  $M \times N$  is the size of the selected region, and  $g(i,j)$  is the intensity.

## Assumption

The selected strip represents the noise characteristics across the entire image.

# Method 2: Estimation from Histogram

## Procedure:

- 1 Compute the histogram of the degraded image
- 2 Fit a known noise model to the histogram
- 3 Estimate parameters using curve fitting or moment matching

## Techniques:

- **Moment Matching:** Match theoretical and sample moments
- **Maximum Likelihood:** Find parameters that maximize likelihood
- **Least Squares Fitting:** Minimize squared error between histogram and model

## Example for Gaussian Noise:

- Sample mean  $\approx$  Population mean  $\mu$
- Sample variance  $\approx$  Population variance  $\sigma^2$

# Comparison of Estimation Methods

| Aspect        | Image Strips            | Histogram             |
|---------------|-------------------------|-----------------------|
| Requirements  | Uniform region needed   | Entire image used     |
| Accuracy      | Good for local noise    | Good for global noise |
| Applicability | Limited regions         | Whole image           |
| Complexity    | Simple computation      | More complex fitting  |
| Best for      | Spatially varying noise | Uniform noise         |

## Practical Tip

Use multiple regions or combine both methods for robust estimation.

# Noise Parameter Estimation - Example

**Given:** An image corrupted by Gaussian noise

**Task:** Estimate  $\mu$  and  $\sigma^2$

## Solution Steps:

- 1 Select a flat region (e.g., sky, uniform background)
- 2 Calculate sample mean:  $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n z_i$
- 3 Calculate sample variance:  $\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (z_i - \hat{\mu})^2$
- 4 Verify by checking if histogram follows Gaussian shape

## Validation:

- Plot estimated PDF over actual histogram
- Check goodness-of-fit using  $\chi^2$  test

## Key Concepts Covered:

- ➊ **Image Restoration:** Objective process to recover original image
- ➋ **Degradation Model:**  $g(x, y) = h(x, y) * f(x, y) + \eta(x, y)$
- ➌ **Noise Models:**
  - Gaussian: Most common, symmetric
  - Rayleigh: Skewed, range imaging
  - Erlang & Exponential: Laser imaging
  - Uniform: Quantization studies
  - Impulse: Salt-and-pepper noise
- ➍ **Parameter Estimation:** From strips or histograms

**Next Topics:** Restoration filters and image compression

## Text Books:

- ① Rafael C. Gonzalez and Richard E. Woods, "*Digital Image Processing*", Pearson Edition, Latest Edition.

## Reference Books:

- ② I. Pitas, "*Digital Image Processing Algorithms*", Prentice Hall, Latest Edition.
- ③ A. K. Jain, "*Fundamental of Digital Image Processing*", Prentice Hall of India Pvt. Ltd., Latest Edition.
- ④ K. Castleman, "*Digital Image Processing*", Prentice Hall of India Pvt. Ltd., Latest Edition.

# Thank You!

Questions?